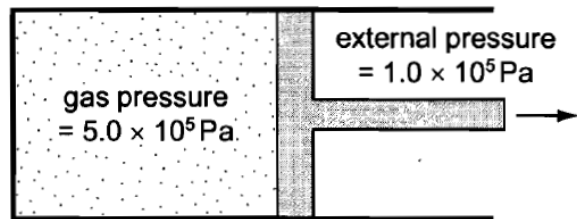


new extension is thus $(2x) \times (3/2) = 3x$.

- 8 A gas at a pressure of $5.0 \times 10^5 \text{ Pa}$ is enclosed in a cylinder fitted with a piston.



The gas expands by 4.0 m^3 against a constant external pressure of $1.0 \times 10^5 \text{ Pa}$.

How much work does the gas do against the external pressure?

- A** $4.0 \times 10^5 \text{ J}$ **B** $12 \times 10^5 \text{ J}$ **C** $16 \times 10^5 \text{ J}$ **D** $20 \times 10^5 \text{ J}$

Ans: A

The gas has to do work against the external pressure. Therefore, the value of the pressure used is the value of the external pressure, not the gas's own pressure.

$$W = p\Delta V = 1.0 \times 10^5 (4.0) = 4.0 \times 10^5 \text{ J}$$

- 9 A ball rolls down the circularly curved track shown below.